WORKED SOLUTION

STEP III 2018, Question 7

Pure • Basel series

STEP III 2018, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s3-q7

This question has a full guided treatment in *Beyond STEP*, Section 2.4, $\pi^2/6$: https://beyondstep.keithmcnulty.org/ch02/basel.html

The key idea

The expression $\cot\theta + \mathrm{i}$ is $(\cos\theta + \mathrm{i}\sin\theta)/\sin\theta$ in disguise, so De Moivre's theorem turns its powers into multiple angles. Expanding the same powers with the binomial theorem gives a polynomial in $\cot^2\theta$, whose roots we can read off from where $\sin(2n+1)\theta$ vanishes. Vieta's formula then sums the roots for free, and squeezing $1/\theta^2$ between $\cot^2\theta$ and $1+\cot^2\theta$ traps the Basel sum between two bounds that tend to the same limit.

Solution

Part (i)

Since $\sin\theta \neq 0$,

$$\cot\theta \pm \mathrm{i} = \frac{\cos\theta \pm \mathrm{i}\sin\theta}{\sin\theta}.$$

By De Moivre's theorem, $(\cos\theta \pm \mathrm{i}\sin\theta)^{2n+1} = \cos(2n+1)\theta \pm \mathrm{i}\sin(2n+1)\theta$, so

$$\frac{(\cot\theta+\mathrm{i})^{2n+1} - (\cot\theta-\mathrm{i})^{2n+1}}{2\mathrm{i}} = \frac{2\mathrm{i}\sin(2n+1)\theta}{2\mathrm{i}\sin^{2n+1}\theta} = \frac{\sin(2n+1)\theta}{\sin^{2n+1}\theta},$$

which is the required identity.

Now expand the left-hand side with the binomial theorem instead. Writing $c = \cot\theta$ and $N = 2n+1$,

$$(c+\mathrm{i})^N - (c-\mathrm{i})^N = \sum_{k=0}^{N} \binom{N}{k} c^{N-k}\,\mathrm{i}^k\left(1-(-1)^k\right).$$

The bracket is 0 for even k and 2 for odd k. Put $k = 2j+1$ with $j = 0, 1, \ldots, n$; then $\mathrm{i}^k = \mathrm{i}(-1)^j$ and $c^{N-k} = c^{2(n-j)}$. Dividing by 2i, and writing $x = \cot^2\theta$,

$$\frac{\sin(2n+1)\theta}{\sin^{2n+1}\theta} = \sum_{j=0}^{n}(-1)^j\binom{2n+1}{2j+1}x^{n-j} = \binom{2n+1}{1}x^n - \binom{2n+1}{3}x^{n-1} + \cdots + (-1)^n.$$

Call this polynomial $P(x)$. It has degree n, with leading coefficient $2n+1 \neq 0$.

Take $\theta_m = \dfrac{m\pi}{2n+1}$ for $m = 1, \ldots, n$. Then $0 < \theta_m < \frac{1}{2}\pi$, so $\sin\theta_m \neq 0$, and $\sin(2n+1)\theta_m = \sin m\pi = 0$. Hence

$$P(\cot^2\theta_m) = 0, \qquad m = 1, 2, \ldots, n.$$

Ask yourself How do we know these are *all* the solutions, and not just some of them?

On $(0, \frac{1}{2}\pi)$, $\cot\theta$ is positive and strictly decreasing, so $\cot^2\theta$ is strictly decreasing too. The angles $\theta_1 < \theta_2 < \cdots < \theta_n$ all lie in this interval, so the n numbers $\cot^2\theta_m$ are distinct. A polynomial of degree n has at most n roots, so these are exactly the solutions of $P(x) = 0$.

Part (ii)

The sum of the roots of $a_n x^n + a_{n-1}x^{n-1} + \cdots$ is $-a_{n-1}/a_n$. Here $a_n = \binom{2n+1}{1}$ and $a_{n-1} = -\binom{2n+1}{3}$, so

$$\sum_{m=1}^{n}\cot^2\left(\frac{m\pi}{2n+1}\right) = \frac{\binom{2n+1}{3}}{\binom{2n+1}{1}} = \frac{(2n+1)(2n)(2n-1)}{6\,(2n+1)} = \frac{n(2n-1)}{3},$$

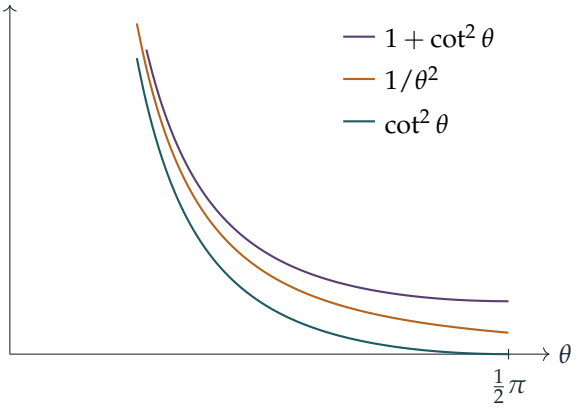
as required.

Part (iii)

For $0 < \theta < \frac{1}{2}\pi$ we are given $0 < \sin\theta < \theta < \tan\theta$. All three are positive, so squaring keeps the order, and taking reciprocals reverses it:

$$\frac{1}{\tan^2\theta} < \frac{1}{\theta^2} < \frac{1}{\sin^2\theta}, \qquad \text{that is} \qquad \cot^2\theta < \frac{1}{\theta^2} < 1 + \cot^2\theta,$$

using $\operatorname{cosec}^2\theta = 1 + \cot^2\theta$.



Put $\theta = \theta_m = \dfrac{m\pi}{2n+1}$, so that $\dfrac{1}{\theta_m^2} = \dfrac{(2n+1)^2}{\pi^2} \cdot \dfrac{1}{m^2}$, and add the inequalities for $m = 1, \ldots, n$. Writing $S_n = \sum_{m=1}^{n} \frac{1}{m^2}$ and using part (ii),

$$\frac{n(2n-1)}{3} < \frac{(2n+1)^2}{\pi^2} S_n < n + \frac{n(2n-1)}{3} = \frac{2n(n+1)}{3}.$$

Multiplying through by $\pi^2/(2n+1)^2 > 0$,

$$\frac{\pi^2}{3} \cdot \frac{n(2n-1)}{(2n+1)^2} < S_n < \frac{\pi^2}{3} \cdot \frac{2n(n+1)}{(2n+1)^2}.$$

Dividing top and bottom by n^2, both fractions tend to $\frac{2}{4} = \frac{1}{2}$ as $n \to \infty$, so both bounds tend to $\frac{\pi^2}{6}$. By the squeeze (sandwich) theorem, $S_n \to \frac{\pi^2}{6}$, which means

$$\sum_{m=1}^{\infty} \frac{1}{m^2} = \frac{\pi^2}{6}.$$

Checks

- For $n = 1$ the polynomial is $3x - 1$, with root $\frac{1}{3} = \cot^2 \frac{\pi}{3}$, and the formula in (ii) gives $1 \times 1/3 = \frac{1}{3}$ too. For $n = 2$, $\cot^2 \frac{\pi}{5} + \cot^2 \frac{2\pi}{5}$ should be 2; numerically $1.894 + 0.106 = 2.000$.
- With $n = 50$ the bounds in (iii) are $1.5964 < S_{50} = 1.6251 < 1.6448$, and $\pi^2/6 \approx 1.6449$. The bounds close in on each other, as they must, since their difference is $\pi^2 n/(2n+1)^2 \approx \pi^2/(4n)$.

Common slips

- Showing that each $\cot^2 \theta_m$ is a root but forgetting to argue that they are distinct, which is what makes them *all* the solutions.

- Getting the sign in Vieta's formula wrong: the coefficient of x^{n-1} is $-\binom{2n+1}{3}$, so the sum of roots is $+\binom{2n+1}{3}/\binom{2n+1}{1}$.
- Reversing the inequalities without noting that all quantities are positive, which is what allows squaring and taking reciprocals.

Beyond STEP

Euler first found $\sum 1/m^2 = \pi^2/6$ in 1734 by treating $\sin x/x$ as an "infinite polynomial" and comparing coefficients, the same Vieta idea used here, but without the rigour that the squeeze supplies. At university you will meet this sum again through Fourier series (Parseval's identity) and as $\zeta(2)$, the first value of the Riemann zeta function; the values $\zeta(2k)$ are all rational multiples of π^{2k}, while very little is known about $\zeta(3)$. For the analysis behind limits and the squeeze theorem, try Abbott, *Understanding Analysis*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.